# Final Held-out Test Report

Neurosymbolic Device Discovery

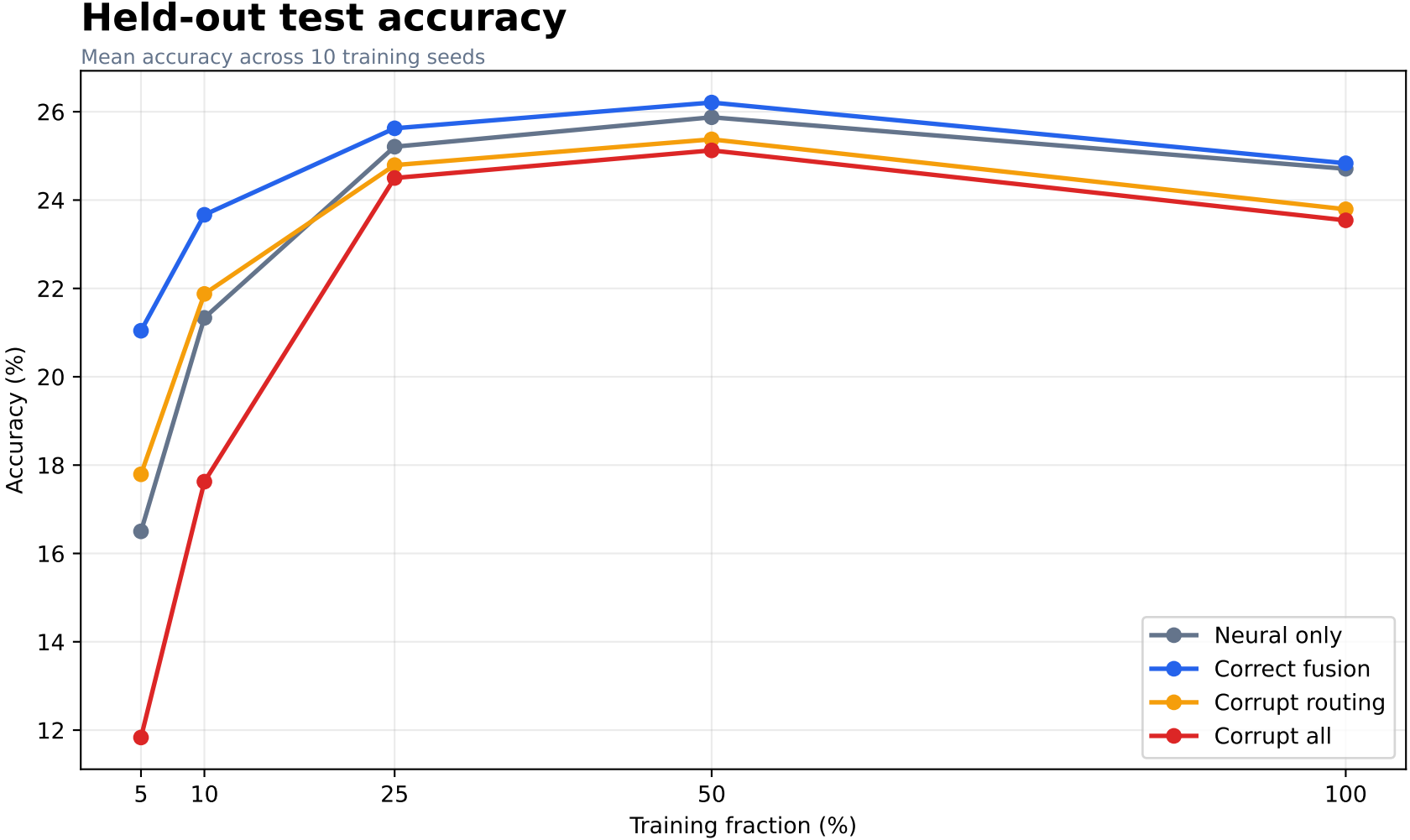
Evaluation setup: 240 held-out synthetic observations, 50 trained MLP checkpoints, 10 seeds per training fraction, and symbolic strength lambda = 0.5.

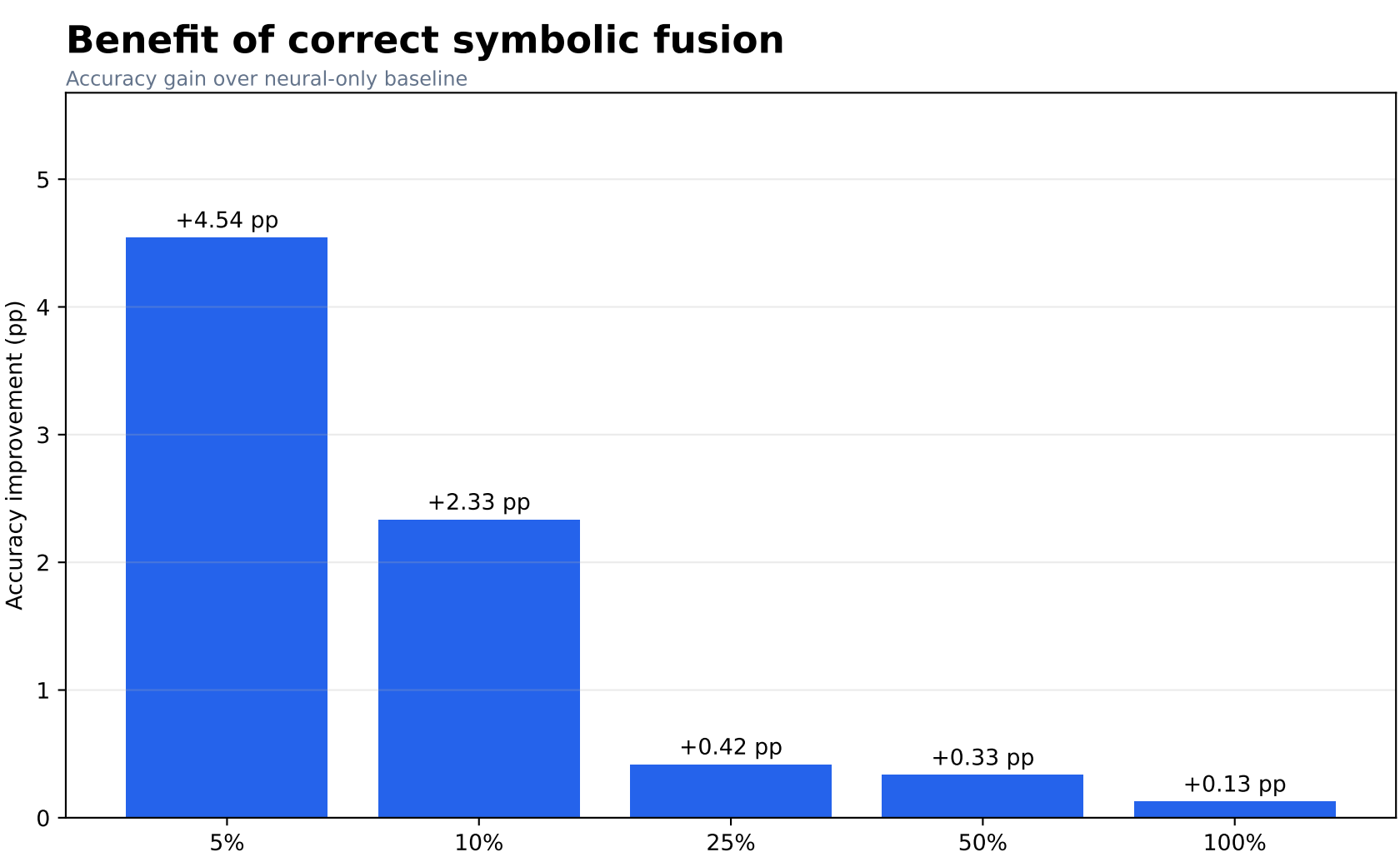
Conditions: neural-only, correct symbolic fusion, corrupted routing rules, and corrupted vendor/routing/VLAN rules.

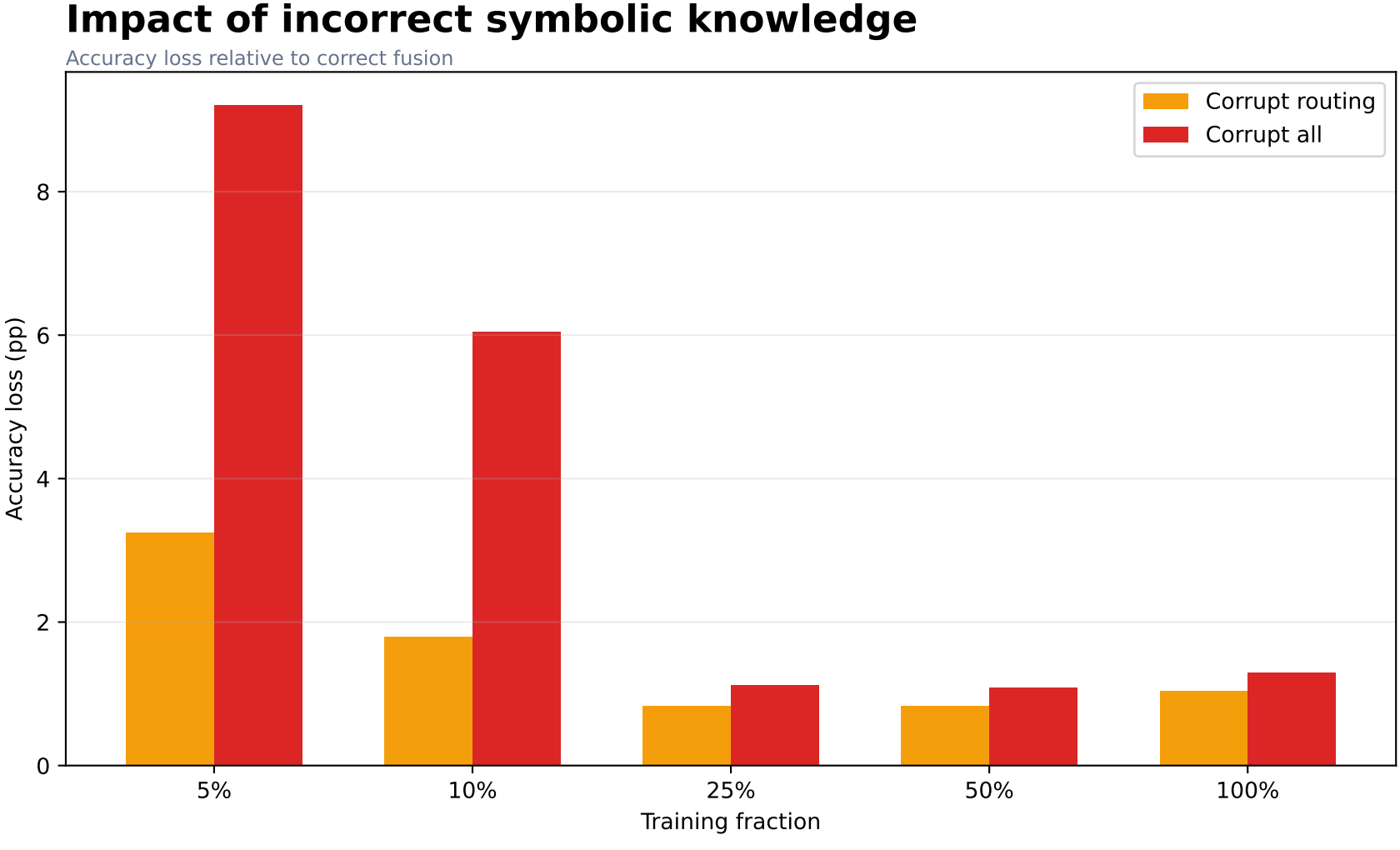
At 5% training data, correct fusion improved accuracy over neural-only by 4.54 percentage points.

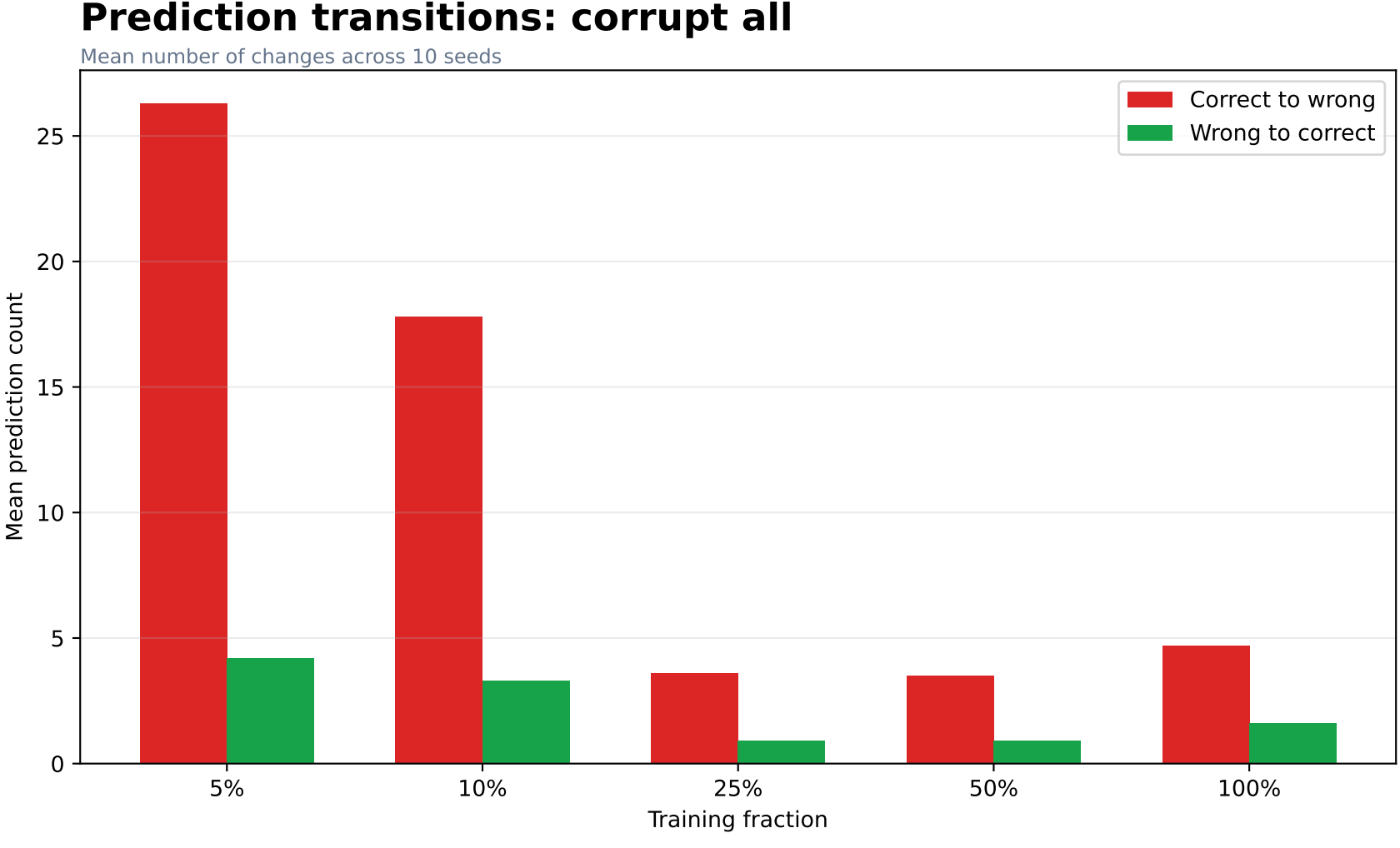
At the same fraction, corrupting all symbolic rules reduced accuracy by 9.21 percentage points relative to correct fusion.

The test set was not used for training or hyperparameter selection in this evaluation.


Held-out test accuracy
Mean accuracy across 10 training seeds
Accuracy (%)
26
24
22
20
18
16
14
12
5
10
25
50
100
Training fraction (%)
Neural only
Correct fusion
Corrupt routing
Corrupt all


Benefit of correct symbolic fusion
Accuracy gain over neural-only baseline
Accuracy improvement (pp)
0
1
2
3
4
5
+4.54 pp
+2.33 pp
+0.42 pp
+0.33 pp
+0.13 pp
5%
10%
25%
50%
100%


Impact of incorrect symbolic knowledge
Accuracy loss relative to correct fusion
Corrupt routing
Corrupt all
Accuracy loss (pp)
0
2
4
6
8
5%
10%
25%
50%
100%
Training fraction


Prediction transitions: corrupt all
Mean number of changes across 10 seeds
Correct to wrong
Wrong to correct
Mean prediction count
0
5
10
15
20
25
5%
10%
25%
50%
100%
Training fraction

# Discussion and limitations

The held-out synthetic test reproduces the main trends observed during validation: symbolic fusion is most helpful with scarce labeled data, and incorrect symbolic knowledge can degrade performance.

The dataset is generated by a simulator rather than collected from real network devices.

All 10 training seeds are evaluated on the same 240 test observations; they are not independent test sets.

The fixed lambda value and symbolic rules were explored during development. The held-out results should therefore be interpreted as a single synthetic test evaluation, not evidence of real-world deployment performance.

Future work should evaluate independent simulator settings, rule reliability, and real network device telemetry.